AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

In the floriculture industry, orchid hybridization is widely used to create new orchid varieties with desirable traits such as flower color, size, shape, fragrance, and blooming performance. However, selecting compatible parent orchids for hybrid pollination still relies heavily on breeder experience and trial-and-error experimentation. This process is time-consuming, unpredictable, and particularly difficult for beginner orchid growers who lack sufficient knowledge of hybrid breeding techniques. Although recent research in orchid cultivation has introduced technologies for disease detection, environmental monitoring, irrigation automation, and species identification, there is currently no intelligent system that predicts hybrid pollination compatibility between parent orchids before the pollination process. This highlights a significant research gap in applying data-driven methods to support orchid hybridization decisions.To address this gap, this research proposes a machine learning–based Hybrid Pollination and Compatibility Analysis system focusing on Vanda orchids, which are widely cultivated in tropical orchid farming but remain underrepresented in computational breeding research. The proposed system integrates computer vision and machine learning techniques to analyze parent orchid characteristics. Images of potential parent orchids are processed using image analysis methods to extract visual features such as color patterns, petal shapes, and textures. These visual features are combined with biological traits of the parent orchids, including species characteristics and blooming attributes, to train a predictive machine learning model capable of estimating hybrid success probability and generating a compatibility score for parent orchid pairs.The expected outcome of this research is an AI-powered decision-support tool that assists orchid breeders and beginner growers in selecting suitable parent orchids before performing hybrid pollination. By transforming traditional experience-based breeding into a data-driven approach, the system aims to reduce trial-and-error breeding, improve hybridization planning, and support more efficient orchid cultivation practices. Furthermore, this research demonstrates the practical application of artificial intelligence in smart agriculture and floriculture.

Keywords: Orchid hybridization, Vanda orchids, hybrid pollination, compatibility prediction, machine learning, computer vision, smart agriculture.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
CNN	Convolutional Neural Network
SVM	Support Vector Machine
SDG	Sustainable Development Goal
RGB	Red Green Blue
VW	Vacin and Went Medium
LKR	Sri Lankan Rupee
SST	Software Systems and Technologies

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1. INTRODUCTION

1.1 Background and Context

Orchids are considered one of the most valuable groups of ornamental plants in the global floriculture industry due to their unique flower shapes, vibrant colours, and long blooming periods. Vanda orchids are considered one of the popular types of orchids among the many types of orchids that are grown across the world due to their large flower sizes, vibrant colours, and hybrid breeding capabilities. Hybridization of orchids is commonly used by plant breeders for the development of new types of orchids with desirable characteristics such as improved flower colours, increased flower sizes, improved fragrances, and better adaptability.

Hybrid pollination is a method of transferring the pollen of an orchid plant onto the stigma of another orchid plant for the purpose of producing a hybrid plant with desirable characteristics from both parents. Hybridization of orchids is a complex and unpredictable phenomenon that is commonly used by plant breeders for the cultivation of orchids. Hybrid pollination of orchids is affected by many biological factors, and the success of hybrid pollination depends on many biological factors. Most plant breeders of orchids rely on their own experience and experimentation while selecting parents for hybridization due to the complex nature of hybridization.

The uncertainty of the outcomes of orchid breeding also presents the following challenges. Orchid hybrids take between three and five years to bloom and display their final flower features. This implies that the breeder has to wait for an extended period before knowing whether the hybridization was successful. Unsuccessful hybridization also implies that the breeder will have wasted considerable time and resources. The challenges of orchid breeding and hybridization apply to various stakeholders in the orchid sector. They include orchid farmers, hobby breeders, researchers in horticulture and floriculture, and the floriculture industry as a whole. In countries like Thailand, Taiwan, and Singapore, orchid cultivation is an important sector in agricultural exports and the ornamental market. There is also an emerging orchid cultivation sector in Sri Lanka that includes small-scale nurseries and hobby breeders of orchids.

The other major challenge is that there is a lack of accessible knowledge and information for a beginner in orchid cultivation. This is where individuals who are interested in orchid breeding do not have adequate information on how hybrid pollination is done. This implies that if individuals do not have adequate information on hybrid pollination, they may not be able to successfully breed hybrid orchids. This, therefore, implies that there is a need for systems that not only help experienced individuals but also provide information for individuals who are undertaking hybrid pollination for the first time.

In recent years, technological advancements in Artificial Intelligence (AI) and Machine Learning (ML) have enabled intelligent decision-support systems in agriculture. These

technologies can analyse complex biological patterns and assist farmers in making informed decisions. Several studies have applied machine learning techniques to orchid cultivation problems. For example, smart farming systems integrates machine learning and IoT technologies to monitor orchid environmental conditions and optimize irrigation, fertilizer application, and disease treatment. Other research has explored the use of deep learning models to detect orchid diseases and classify orchid species using plant images.

While the above solutions illustrate the importance of AI in the cultivation of orchids, most existing AI models are geared towards the monitoring of plant health, irrigation systems, and identification of plant species. However, very few models have been proposed to tackle the challenge of predicting the compatibility of the pollination of hybrids among the parent orchids. This means that breeders still do not have intelligent systems to analyze the characteristics of the parent orchids to estimate the chances of successful breeding.

In order to address this limitation, this research proposes a Hybrid Pollination and Compatibility Analysis system that utilizes machine learning algorithms for orchid breeding decision support. The proposed system will evaluate the traits of the parent orchids based on their species type, blooming characteristics, and flower morphology, along with visual features extracted from the orchid image. Using these features, the proposed system will be able to determine the likelihood of successful hybridization between orchid species and provide a compatibility score for orchid breeding decision support. Moreover, the proposed system will be able to provide guidance on hybrid pollination techniques for beginner orchid growers.

From a technology point of view, the solution that is being proposed is an intelligent system that incorporates machine learning technology, image feature extraction technology, and predictive analytics technology. Hence, this study fits well in the Software Systems and Technologies (SST) cluster of research studies that emphasize the design and development of sophisticated software solutions that can solve real-world problems by utilizing modern computing technologies. The integration of orchid biological knowledge and machine learning technology in a software platform is an exemplification of how software technology can be harnessed for the development of innovation in the field of agriculture.

Furthermore, this research contributes to Sustainable Development Goal (SDG) 9 – Industry, Innovation, and Infrastructure, which promotes technological innovation to improve productivity and support sustainable industries. By introducing an AI-assisted system for orchid hybrid compatibility prediction, the research supports the modernization of floriculture practices and encourages the adoption of intelligent technologies in horticultural production.

1.2 Literature Review

The proposed solution is an AI-powered Smart Orchid Care System that integrates IoT sensors, computer vision, and multi-modal machine learning to support the complete orchid cultivation lifecycle. The system is designed to overcome the limitations of existing orchid

care solutions by combining disease intelligence, smart watering and nutrient forecasting, growth-stage prediction, and hybrid pollination analysis into a single unified platform.

Recent studies in orchid breeding and plant biotechnology highlight the complexity of hybrid pollination and the importance of compatibility between parent plants. Orchid hybridization success depends on multiple biological factors such as genetic traits, pollen–pistil interactions, and environmental conditions. Research conducted on Dendrobium orchids demonstrated that only certain species combinations successfully produced fruits after pollination due to compatibility differences between male and female reproductive structures, while other combinations failed because of genetic or physiological incompatibility. These findings indicate that orchid breeders often rely on trial-and-error methods when selecting parent plants for hybridization, which can be time-consuming since orchids may require several years before the success of a hybrid cross can be confirmed. In addition to biological research, recent agricultural studies have explored technological approaches to support pollination and breeding processes, including automated pollination systems and advanced breeding technologies designed to improve plant productivity and address pollination challenges in agriculture [1] [2].

Orchid breeding research has also explored genetic improvement techniques such as colchicine treatment to enhance hybrid characteristics and improve orchid varieties through in vitro breeding approaches [3]. These approaches demonstrate how biological breeding techniques can enhance orchid hybridization outcomes, but they still require extensive experimentation and long evaluation periods before successful hybrids can be confirmed.

Alongside these developments, machine learning and deep learning techniques have been widely applied in plant image analysis and botanical classification tasks. Recent studies have shown that convolutional neural networks (CNNs) and transfer learning models such as ResNet can accurately classify flower species by extracting visual features such as petal shape, color patterns, and morphological characteristics from plant images [4] [5]. Systems such as Deep Flower and automated flower recognition frameworks demonstrate how deep learning models can achieve high accuracy in flower classification and botanical identification tasks [6] [7]. Furthermore, artificial intelligence has been applied in agricultural research to analyze plant traits, optimize plant breeding strategies, and support phenotyping processes through data-driven approaches [8] [9].

In addition to AI-based plant analysis, several studies have investigated the use of embedded systems, sensors, and intelligent environmental monitoring for orchid cultivation. Sensor-based monitoring systems can collect environmental data such as temperature and humidity

and transmit it to external servers for monitoring and control, enabling better environmental management in orchid nurseries and improving breeding success rates [10].

However, most existing systems primarily focus on flower species recognition, plant disease detection, or environmental monitoring rather than predicting compatibility between parent plants for hybrid breeding. Consequently, breeders still lack intelligent decision-support systems capable of analyzing orchid traits and estimating the probability of successful hybrid pollination before performing the cross. This limitation highlights the need for an intelligent Hybrid Pollination and Compatibility Analysis system that integrates plant trait analysis and machine learning techniques to assist orchid breeders in selecting suitable parent plants.

1.3 Research Gap

Although several studies have investigated orchid breeding and plant classification using machine learning techniques, many limitations still exist in current research related to hybrid pollination prediction. Orchid hybridization is a complex biological process that depends on genetic compatibility, flower morphology, blooming time, and physiological interactions between pollen and pistil. Research on *Dendrobium* orchids shows that only certain species combinations successfully produce fruits after pollination, while others fail due to incompatibility between reproductive structures. As a result, orchid breeders typically rely on trial-and-error approaches when selecting parent plants for hybridization. This process is inefficient because orchids may take several years to bloom, meaning breeders must wait a long time before determining whether a hybrid cross has been successful.

Another limitation in existing research is the lack of predictive decision-support systems for orchid hybridization. While recent studies have applied artificial intelligence and deep learning techniques for plant image recognition and flower classification, these systems primarily focus on identifying plant species or detecting plant diseases rather than assisting plant breeding decisions [4] [5]. Deep learning models such as convolutional neural networks (CNNs) and transfer learning architectures have demonstrated high accuracy in flower classification by analysing visual features such as petal structure, colour patterns, and morphological characteristics [6] [9]. However, these approaches do not analyse compatibility between parent plants or predict the probability of successful hybrid pollination.

Furthermore, many existing agricultural AI systems analyse only limited plant traits or environmental conditions without integrating biological breeding knowledge. Hybridization compatibility depends on multiple factors including flower morphology, species characteristics, blooming stages, and genetic traits. Systems that focus solely on species identification or environmental monitoring cannot effectively support orchid breeders in selecting suitable parent combinations for hybridization. Consequently, there is still a lack of

intelligent systems that combine image-based plant trait analysis with predictive machine learning models to estimate hybrid pollination compatibility.

To address these limitations, this research proposes a Hybrid Pollination and Compatibility Analysis System for Vanda Orchids using Machine Learning and Image Feature Extraction.

The main research contributions of this study are:

• Development of an intelligent compatibility analysis system that evaluates hybrid pollination potential between parent orchids.

• Use of flower image analysis and machine learning techniques to extract morphological traits such as petal structure, colour patterns, and flower shape.

• Implementation of predictive models that estimate the probability of successful hybridization before pollination is performed.

• Integration of biological breeding knowledge and image-based plant trait analysis to support decision-making in orchid hybridization.

• Focus on **Vanda orchids**, which are widely cultivated for hybrid breeding but have limited technological support for compatibility prediction.

By integrating machine learning, image feature extraction, and orchid breeding knowledge, the proposed system aims to reduce uncertainty in hybrid pollination and assist orchid breeders in selecting suitable parent combinations.

Table 1: : Research Gap Comparison

Research Paper	Approach	Technologies Used	Limitations
Pollination compatibility of *Dendrobium* orchids from Balij [1]	Experimental study analysing pollination success between orchid species	Controlled manual pollination experiments, **in-vitro seed culture using Vacin & Went (VW) medium**, statistical analysis (ANOVA)	Focuses on biological pollination experiments only. Does not provide predictive models or intelligent systems to estimate hybrid compatibility before pollination.
DeepFlower: A Deep Learning Approach for Accurate Flower Classification [4]	DeepFlower: A Deep learning based flower classification system	Convolutional Neural Networks (CNN), TensorFlow/Keras deep learning framework, image dataset training	Designed for general flower classification. Does not analyse orchid-specific traits or hybrid pollination compatibility.
Transformer Network-Based Image Segmentation Using Hybrid [6]Flower	Transfer learning model for flower image recognition	ResNet50 architecture, transfer learning, PyTorch deep learning framework	Identifies flower species but does not support plant breeding decisions or hybrid compatibility prediction.

Pollination Optimization			
Breeding of ornamental orchids with focus on Phalaenopsis: current approaches, tools, and challenges for this century [8]	Review of modern orchid breeding techniques, hybridization strategies, and biotechnology tools used in ornamental orchid improvement	Genetic breeding techniques, molecular markers, tissue culture propagation, biotechnology tools for orchid breeding	Focuses on biological breeding strategies and biotechnology approaches. Does not provide AI-based systems or predictive models to estimate hybrid pollination compatibility.
Mobile Nursery Construction with Alignment of Sensors for Orchids Breeding [10]	Sensor-based environmental monitoring system for orchid breeding nurseries	Temperature and humidity sensors, embedded ARM system, environmental monitoring system, data communication with external server	Focuses on environmental monitoring of orchid nurseries. Does not analyse flower morphology or predict hybrid pollination compatibility between orchid species.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop a machine learning–based Hybrid Pollination and Compatibility Analysis system that predicts the likelihood of successful hybridization between Vanda orchid parent plants and provides pollination guidance to assist breeders in selecting suitable parent combinations and performing hybrid pollination correctly within the proposed research period.

2.2 Specific Objectives

These follow a logical research workflow from start → system output.

- To collect and prepare a dataset of Vanda orchid flower images and relevant parent plant traits required for hybrid pollination compatibility analysis.
- To preprocess orchid flower images and extract important morphological features such as petal shape, color patterns, and structural characteristics using image processing techniques.
- To analyse relationships between orchid parent plant traits and hybridization outcomes in order to identify factors influencing pollination compatibility.
- To design and train a machine learning model capable of predicting the probability of successful hybrid pollination between selected parent orchids.
- To develop a compatibility analysis module that generates compatibility scores and recommends suitable parent orchid combinations for hybrid breeding.
- To design and implement a pollination guidance module that provides step-by-step instructions and recommendations to assist users in performing hybrid pollination correctly.
- To evaluate the performance and usability of the proposed system through prediction accuracy testing and user validation.

3. METHODOLOGY

- **Research Approach**

  - This research follows a Design Science Research approach combined with Experimental validation.
  - The Design Science approach is used to design and develop a technological solution that addresses the problem of uncertainty in orchid hybrid pollination. In this research, the proposed artifact is a Hybrid Pollination and Compatibility Analysis System for Vanda orchids that predicts hybrid compatibility and provides pollination guidance using machine learning and image analysis techniques.
  - The Experimental approach is used to evaluate the performance of the proposed system. Image data collected from orchid flowers and plant trait information will be used to train and test machine learning models. The system will be experimentally evaluated by measuring prediction accuracy and system usability.
  - The overall research process includes:
    - dataset collection of orchid images and plant traits
    - image preprocessing and feature extraction
    - machine learning model development
    - compatibility prediction system implementation
    - pollination guidance module development
    - system validation and performance evaluation

- **Technical Methods and Algorithms**

  - The proposed system uses computer vision and machine learning techniques to analyse orchid flower traits and predict hybrid compatibility.

  - Image Processing
    - Flower images captured using a camera will be preprocessed using image processing techniques including:
      - image resizing
      - noise reduction
      - image normalization
      - segmentation of flower regions

    - These processes help extract meaningful features from orchid flowers.

  - Feature Extraction
    - Morphological characteristics of orchid flowers will be extracted from images, including:
      - petal shape
      - flower symmetry

- color patterns
- lip structure (labellum)
- flower size and structure

- These features represent biological traits used by breeders when selecting hybrid parents.

- Machine Learning Models
  - The extracted features and plant trait information will be used as input to machine learning models that predict hybrid pollination compatibility.
  - Several machine learning algorithms will be evaluated, including:
    - Random Forest
    - Support Vector Machine (SVM)
    - Logistic Regression
  - These models will classify hybrid combinations into:
    - Compatible
    - Low Compatibility
  - The best-performing model will be selected based on prediction accuracy.

- Pollination Guidance Module
  - Based on compatibility predictions and biological knowledge, the system will generate pollination guidance including:
    - recommended parent orchid combinations
    - step-by-step hybrid pollination instructions
    - recommended pollination timing based on flowering stages
  - This module helps beginner growers perform hybrid pollination correctly.

- **Tools and Platforms**

- The following hardware and software tools will be used to implement the system.
  - Hardware
    - Raspberry Pi Camera Module

      A camera will capture high-resolution images of orchid flowers for feature extraction and compatibility analysis.
  - Environmental Sensors
    - Sensors will collect environmental parameters that may influence flowering and pollination conditions.

    - Examples include:
      - DHT22 temperature and humidity sensor
  - Software
    - Python

      Python will be used as the main programming language due to its strong support for machine learning and image processing.
  - Machine Learning Libraries

- ❖ Several open-source libraries will be used, including:
  - • Scikit-learn for machine learning models
  - • NumPy and Pandas for data processing
  - • TensorFlow or PyTorch for deep learning experiments

- Image Processing Framework
  - ❖ OpenCV will be used for image preprocessing and feature extraction tasks.
- Database System
  - ❖ Firebase or a cloud-based database will be used to store collected image datasets, compatibility results, and system records.

- **Data Collection and Ethical Considerations**

  - The dataset for this research will consist of orchid flower images and plant trait information.
  - Images of Vanda orchid flowers will be collected using a camera from orchid plants located in greenhouse environments or orchid farms. Images will be captured under controlled lighting conditions and at different flowering stages.
  - The dataset will include:
    - • orchid flower images
    - • species type
    - • flower morphology characteristics
    - • parent orchid combinations used in hybridization
    - • pollination outcomes (successful / unsuccessful)
  - Environmental parameters such as temperature, humidity, and light intensity may also be recorded to provide additional contextual data.
  - Images will be labeled with the assistance of orchid growers or domain experts to ensure accuracy.
  - Since the study does not involve personal or sensitive human data, ethical risks are minimal. However, proper data management practices will be followed to ensure that all collected data are used only for research purposes.

- **Validation Strategy and Performance Metrics**

  - The collected dataset will be divided into three subsets:
    - • 70% training dataset
    - • 15% validation dataset
    - • 15% testing dataset
  - Cross-validation techniques will be used to ensure model reliability.
  - The performance of the machine learning model will be evaluated using standard classification metrics including:
    - • Accuracy – proportion of correct compatibility predictions
    - • Precision – correctness of predicted compatible hybrid combinations
    - • Recall – ability of the model to correctly detect compatible hybrids
    - • F1 Score – balance between precision and recall

- A target accuracy of greater than 80% is expected for the compatibility prediction model.
- Additionally, the usability of the pollination guidance module will be evaluated by comparing system recommendations with expert breeder feedback.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

- **Overview of the Proposed Solution and Its Components**

The proposed system is an AI-powered smart orchid cultivation platform designed to support growers in managing orchid plants using machine learning and data-driven decision making. The system integrates multiple intelligent components that assist with plant health monitoring, environmental management, and hybrid breeding.
The system consists of four main components, each responsible for a specific aspect of orchid cultivation.

1. Orchid Disease Detection and Treatment Recommendation
This component analyzes orchid leaf and plant images to detect diseases using computer vision techniques. Based on the identified symptoms, the system recommends appropriate treatment methods to help growers maintain plant health.

2. Orchid Growth Stage Recognition and Bloom Prediction
This component identifies orchid growth stages from images and predicts flowering periods using environmental data and machine learning models. This helps growers monitor plant development and anticipate blooming times.

3. Smart Watering and Automated Fertilization
This component monitors environmental conditions such as temperature, humidity, light intensity, and moisture using IoT sensors. Machine learning models analyze this data to determine watering needs and automatically activate irrigation or fertilization systems.

4. Hybrid Pollination and Compatibility Analysis
This component assists orchid breeders by predicting compatibility between two orchid parents. It analyzes flower images and plant traits using machine learning models to estimate hybrid success probability and provide pollination guidance.
Together, these components form an integrated intelligent orchid management system that supports plant health monitoring, environmental control, and hybrid breeding decision making.

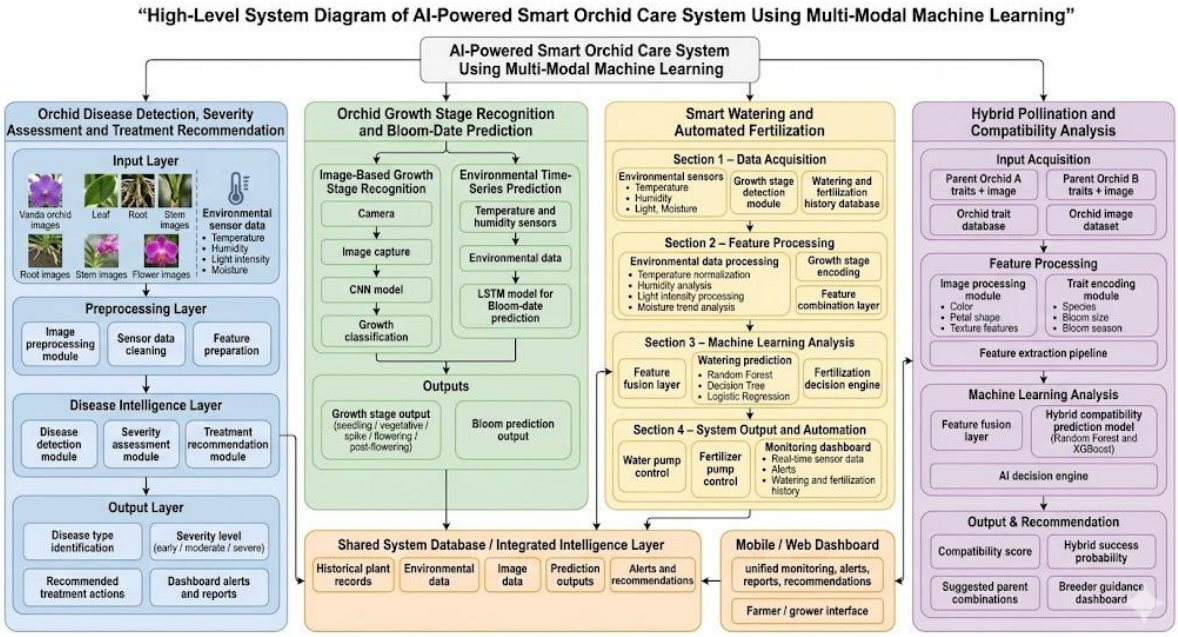

Figure 1:High-Level System Diagram of AI-Powered Smart Orchid Care System Using Multi-Modal Machine Learning

- **Focused Component : Hybrid pollination and compatibility Analysis**

  The Hybrid Pollination and Compatibility Analysis component supports orchid breeders in selecting suitable parent orchids for hybridization.
  Hybrid pollination traditionally relies on breeder experience and manual experimentation. This component improves the process by using image processing and machine learning techniques to analyze orchid flower characteristics and plant traits. The component consists of several main modules.

  Input Acquisition
  The system collects orchid flower images and plant trait information such as species, bloom size and flowering season. These inputs are used for compatibility analysis.

  Image Processing
  Captured orchid images are processed using computer vision techniques to extract features such as color patterns, petal shape, and texture.

  Trait Encoding
  Plant trait information is converted into structured data so that it can be combined with image features for machine learning analysis.

  Machine Learning Compatibility Model
  The extracted image features and encoded plant traits are used as inputs to a machine learning model such as **Random Forest or XGBoost**, which predicts compatibility between two orchid parents.

Decision and Recommendation Module
Based on the prediction results, the system generates outputs including compatibility scores, hybrid success probability, recommended parent combinations, and pollination guidance.
These results are displayed through a breeder guidance dashboard, helping growers perform hybrid pollination more effectively.

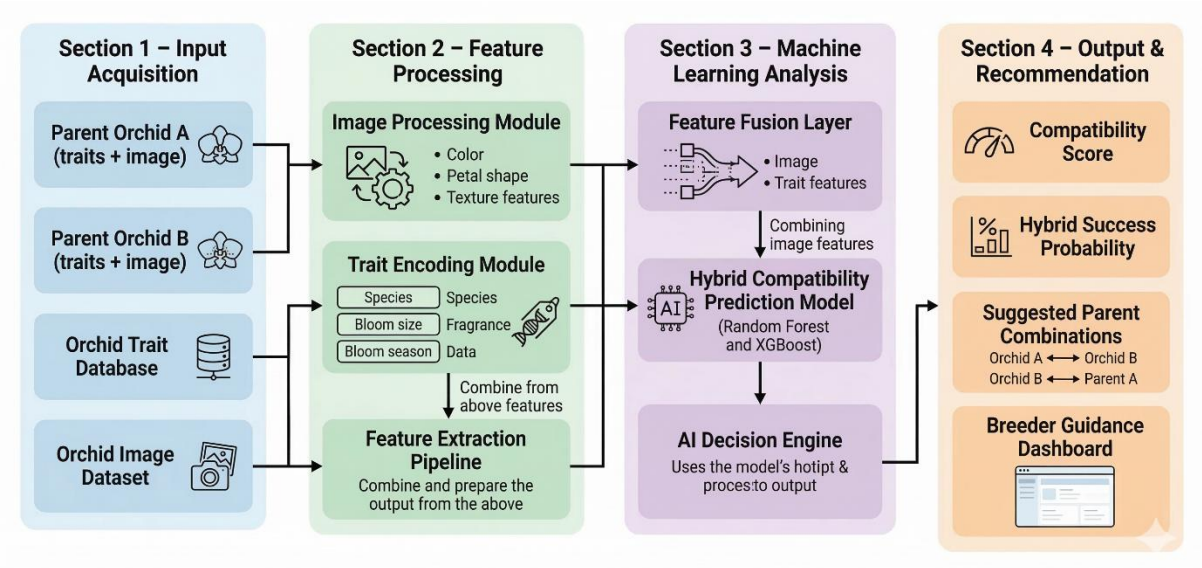

Figure 2:Hybrid pollination and compatibility component overview

- **Component Interactions**

The components of the proposed system interact with each other through shared data and environmental monitoring.

For example, the growth stage recognition component can provide information about the flowering stage of orchids. This information is useful for the hybrid pollination component because pollination should only occur during appropriate flowering stages. Similarly, the smart watering and fertilization component helps maintain optimal environmental conditions that support healthy plant growth and flowering. A stable growing environment improves the success of hybrid breeding experiments.

The disease detection component also supports the system by ensuring that orchids used for hybrid breeding are healthy and free from infections that could affect pollination results.

Through these interactions, the components collectively support orchid cultivation, plant health management, and hybrid breeding decision making.

- **Data Flow Explanation**

  The data flow for the Hybrid Pollination and Compatibility Analysis component starts with the process of orchid parent data collection, which involves the collection of flower images along with the orchid plant's traits, including morphological features and genetic factors. The collected features are then processed in the feature fusion engine, which normalizes the features to produce a structured dataset.
  The combined features are then processed in the compatibility knowledge base, which stores historical hybridization records along with orchid plant traits. The combined features are then processed in the hybrid prediction model, which uses machine learning to predict the compatibility between two orchid parents.
  The results are then processed in the pollination guidance system, which provides compatibility scores along with guidance for the hybridization process through the dashboard.

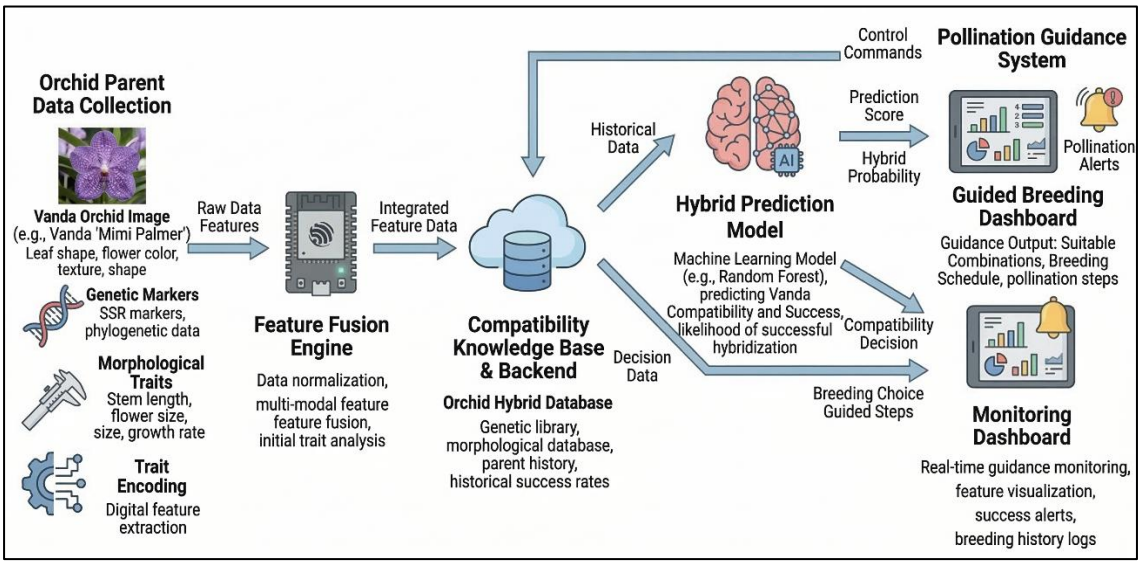

Figure 3: Data Flow Explanation

5. USER REQUIREMENTS

- **Stakeholder Identification**

Several stakeholders interact with the proposed AI-powered orchid cultivation system. Each stakeholder benefits from different system capabilities related to plant monitoring, hybrid breeding, and cultivation support.

Orchid Growers

Orchid growers are the primary users of the system. They use the platform to monitor plant conditions, receive cultivation recommendations, and obtain guidance for hybrid pollination. The system helps growers select suitable parent orchids and improve breeding success.

Orchid Breeders

Orchid breeders use the hybrid compatibility analysis component to identify potential parent combinations for hybridization. The system provides compatibility predictions and pollination guidance that support more efficient breeding experiments.

Home Gardeners

Individuals who grow orchids at home can use the system to understand plant conditions and obtain recommendations for orchid care. The hybrid compatibility feature also helps hobby growers experiment with hybrid breeding.

Agricultural Researchers

Researchers and agricultural experts may use the collected orchid image datasets and plant trait information to study orchid growth patterns, hybrid breeding behavior, and machine learning applications in agriculture.

System Administrators

System administrators maintain the system infrastructure, manage databases, ensure proper functioning of the platform, and monitor system performance and security.

- **Functional Requirements**

  FR1: The system shall capture orchid flower images automatically using a fixed camera installed in the orchid growing environment.
  FR2: The system shall analyze captured images to detect orchid flowering stages.
  FR3: The system shall identify flowers that are ready for pollination.
  FR4: The system shall notify users when orchids enter the pollination-ready stage.
  FR5: The system shall allow users to select two orchid plants for hybrid pollination analysis.

FR6: The system shall analyze flower images and plant traits to predict hybrid compatibility.
FR7: The system shall generate a compatibility score and hybrid success probability.
FR8: The system shall provide step-by-step guidance for performing hybrid pollination.
FR9: The system shall store hybrid pollination records for future analysis.

- **Non-Functional Requirements**

Non-functional requirements describe system performance, reliability, usability, and security expectations.

Performance Requirements
NFR1: The system should process orchid flower images and generate compatibility predictions within 5 seconds.
NFR2: The system should handle multiple compatibility prediction requests without significant delay.

Reliability Requirements
NFR3: The system should operate continuously and maintain stable processing of orchid image data.
NFR4: The system should ensure reliable communication between image capture devices, backend processing modules, and the database.

Usability Requirements
NFR5: The system interface should be simple and easy to understand for both expert breeders and beginner orchid growers.
NFR6: The dashboard should clearly display compatibility scores, predictions, and pollination recommendations.

Security Requirements
NFR7: Access to the system dashboard should be protected through authentication mechanisms.
NFR8: Orchid image datasets and plant trait information stored in the database should be protected from unauthorized access.

Scalability Requirements
NFR9: The system should support expansion to include additional orchid species and larger image datasets in the future.
NFR10: The system architecture should allow integration of additional machine learning models or new breeding analysis modules.

6. COMMERCIALIZATION PLAN

1. Target Market and Customer Persona

The proposed **Hybrid Pollination and Compatibility Analysis system** is designed mainly for orchid breeders, orchid farms, and plant researchers who are interested in improving hybrid breeding success using data-driven analysis.

The main target users are:

Orchid Breeding Farms

Commercial orchid farms that develop new orchid hybrids for sale can use the system to analyze compatibility between parent orchids and improve the success rate of hybrid breeding. This reduces trial-and-error breeding experiments and saves time and resources.

Orchid Hobby Breeders

Home orchid enthusiasts who experiment with hybrid pollination can use the system to obtain compatibility predictions and pollination guidance without requiring extensive botanical expertise.

Agricultural and Botanical Researchers

Researchers studying orchid hybridization, plant genetics, and smart agriculture technologies can use the collected hybridization data and analysis results for research and experimentation.

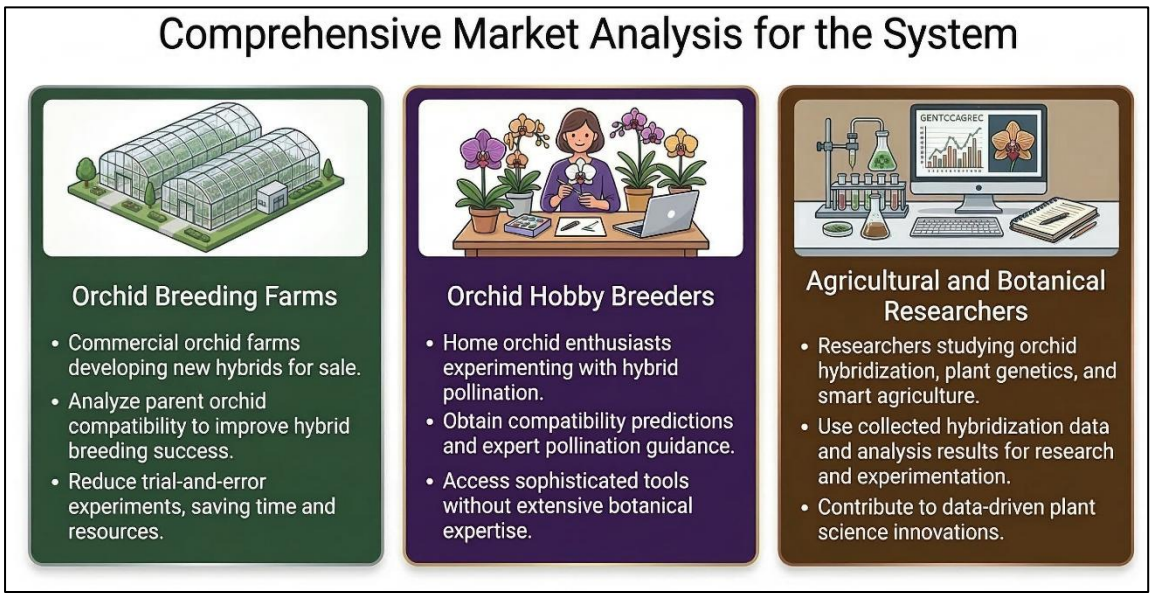

Figure 4: Target Market and Customer Persona

2. Value Proposition

The proposed system provides several benefits to orchid growers and breeders:

• Predicts compatibility between orchid parents before performing hybrid pollination
• Reduces trial-and-error breeding experiments
• Provides step-by-step pollination guidance for growers
• Uses image analysis and plant traits to support data-driven breeding decisions
• Stores hybridization history and breeding records for future analysis
• Supports smarter orchid breeding compared with traditional manual methods

Compared with traditional hybridization methods that rely heavily on breeder experience, the proposed system introduces AI-assisted breeding decision support.

3. Revenue Model

The proposed smart orchid care system can generate revenue through a combination of hardware deployment, intelligent software services, and agricultural decision-support tools.

Smart Orchid Monitoring Device

The system can be offered as an integrated hardware solution installed in orchid farms or greenhouses. The device may include cameras, environmental sensors, and a microcontroller that continuously collects plant images and environmental data. Farms and nurseries can purchase this device to automate monitoring and data collection for orchid cultivation.

AI-Based Plant Analysis Platform

The collected data can be processed through an AI-powered platform that provides multiple analytical services, including disease detection, growth stage recognition, bloom prediction, smart watering recommendations, and hybrid compatibility analysis. Access to this platform can be provided through a subscription-based web or mobile dashboard.

Professional Cultivation Insights

Commercial orchid farms may benefit from advanced cultivation insights generated from historical environmental data, plant growth patterns, and hybrid breeding results. These insights can help farms improve plant health, optimize watering schedules, and identify successful breeding strategies.

Technology Licensing

In the future, the developed AI models and monitoring platform could be licensed to agricultural technology companies, greenhouse automation providers, or orchid nurseries that want to integrate smart plant monitoring capabilities into their existing systems.

4. Cost Estimation

The hybrid compatibility system mainly requires image capture hardware and computing resources for machine learning processing.

Table 2: Cost Estimation

Item	Estimated Cost (LKR)	Description
Raspberry Pi Camera Module	4500 – 6000	Captures orchid flower images
Raspberry Pi / Edge Device	*3500 – 4500*	*Image processing and system control*
ESP32 / Microcontroller	*2000 – 3000*	*Device communication and data transfer*
Power Supply & Cables	*2000 – 3000*	*Hardware connectivity*
Cloud Storage / Database	*Minimal (free tier initially)*	*Stores image datasets and prediction results*
Software Tools	*0*	*Python, OpenCV, Scikit-learn (open source)*

5. Pricing Strategy

The system can use a **hybrid pricing model combining software access and analytics services**.

Table 3: Pricing Strategy

Plan	Monthly Price (LKR)	Features
Basic Plan	500	Hybrid compatibility prediction and pollination guidance
Premium Plan	1000	Advanced hybrid analytics, breeding history tracking, and detailed compatibility insights

6. Competitive Advantage

The proposed system offers several advantages compared with existing orchid breeding practices.

• Uses machine learning to predict hybrid compatibility before performing pollination
• Combines image analysis with plant trait data for more accurate predictions
• Provides pollination guidance to support beginner growers
• Stores historical hybridization data for long-term breeding analysis
• Designed specifically for orchid breeding rather than general plant monitoring systems

Most existing orchid breeding methods rely on manual experimentation and breeder experience. The proposed system introduces AI-assisted hybrid breeding decision support, which can improve efficiency and reduce unsuccessful hybridization attempts.

7. Intellectual Property Considerations

The proposed system involves several elements that may require intellectual property protection.

• Machine learning models developed for orchid hybrid compatibility prediction
• Orchid hybridization datasets and training data collected during the research
• Software platform design and system architecture

These components could potentially be protected through **software copyrights or intellectual property registration** if the system is commercialized in the future.

7. BUDGET AND JUSTIFICATION

The following table shows the estimated budget required to develop the AI Powered Smart Orchid Care System Using Multi-Modal ML.

Table 4: BUDGET AND JUSTIFICATION

Hardware Costs (per unit)			
Cost Category	Estimated Amount (LKR)	Notes	Justification
Raspberry Pi Camera Module	4,500–6,000	Captures plant images	Captures daily images of Vanda orchids to track growth stages (seedling, vegetative, spike, flower, post-bloom).
DHT22 Temperature/Humidity Sensor	900–1,200	Measures environment	Measures greenhouse environment. Temperature and humidity are key factors affecting flowering time and must be included in the prediction model.
ESP32 Microcontroller	2,000–3,000	Processes sensor data	Reads sensor data and sends it to the cloud via WiFi. Low cost and low power consumption.
Raspberry Pi Zero 2W	3,500–4,500	Runs image processing	Runs the image processing at the edge. It captures and analyzes images before sending results to the cloud.
Power supply, cables, SD card	2,500–3,500	Basic components	Provides stable power to the Raspberry Pi and sensors. Stores the operating system and temporary image data. Connects sensors to the board.
3D printed enclosure	1,000–1,500	Protects electronics	Protects electronics from greenhouse humidity, dust, and heat.
Total Hardware per Unit	14,400–19,700		
Software Costs			
Cost Category	Estimated Amount (LKR)	Notes	Justification
All software (Python, TensorFlow, etc.)	0	Open source	Free, open-source
Total Software	0		
Cloud Services (monthly)			
Cost Category	Estimated Amount (LKR)	Notes	Justification

Firebase free tier	0	Up to 1 GB storage	Provides database and storage for free up to 1GB, which is enough for a pilot project with one unit.
Total Cloud (Year 1)	0		
Human Effort Costs			
Cost Category	Estimated Amount (LKR)	Notes	Justification
Travel to greenhouses (fuel, bus fare)	5,000–8,000	10–15 visits	
Mobile data/Wi-Fi charges	5,000–8,000	For cloud testing and updates	
Refreshments during fieldwork	4,000–6,000	Snacks/meals during long days	
Total Human Effort	14,000–22,000		

8. WORK BREAKDOWN STRUCTURE (WBS)

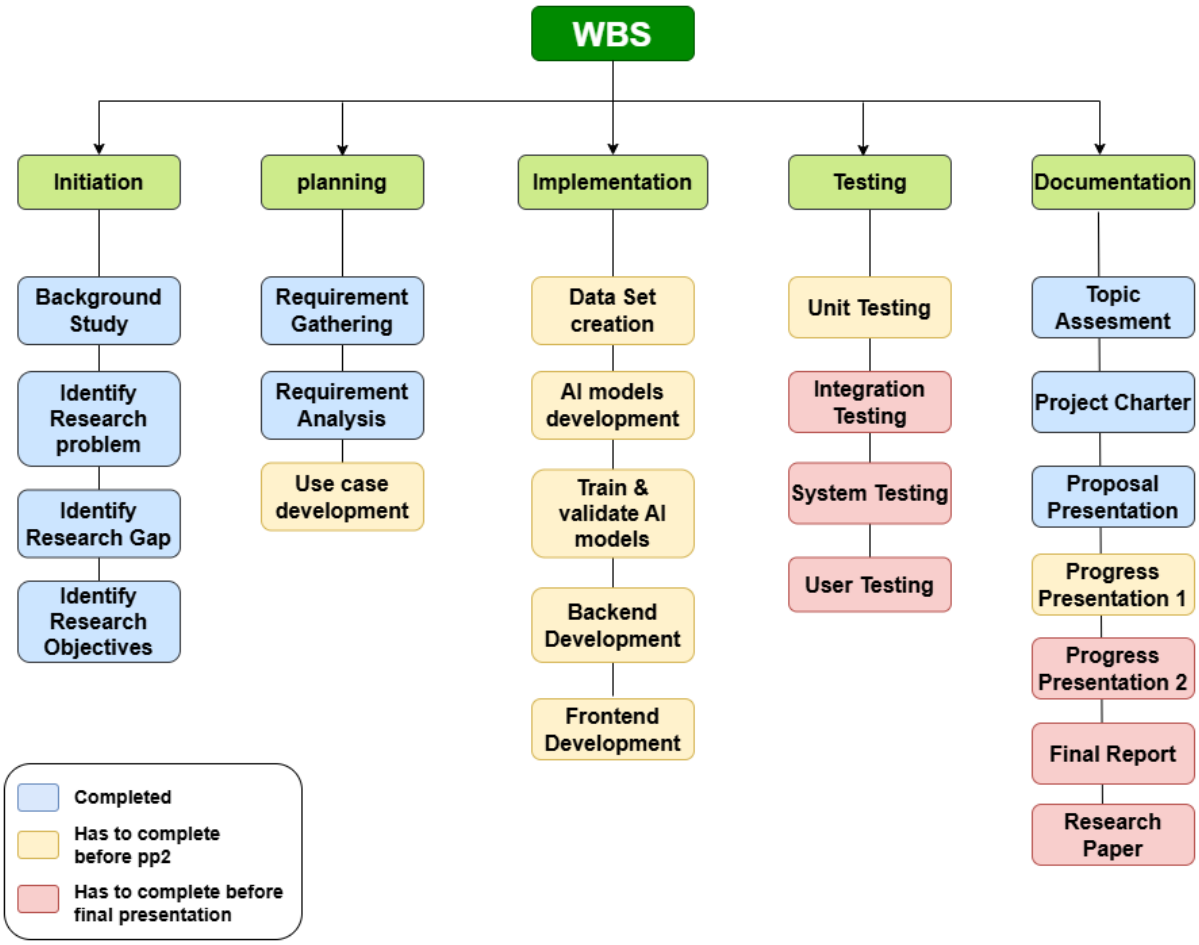

Figure 5: WORK BREAKDOWN STRUCTURE

9. GANTT CHART

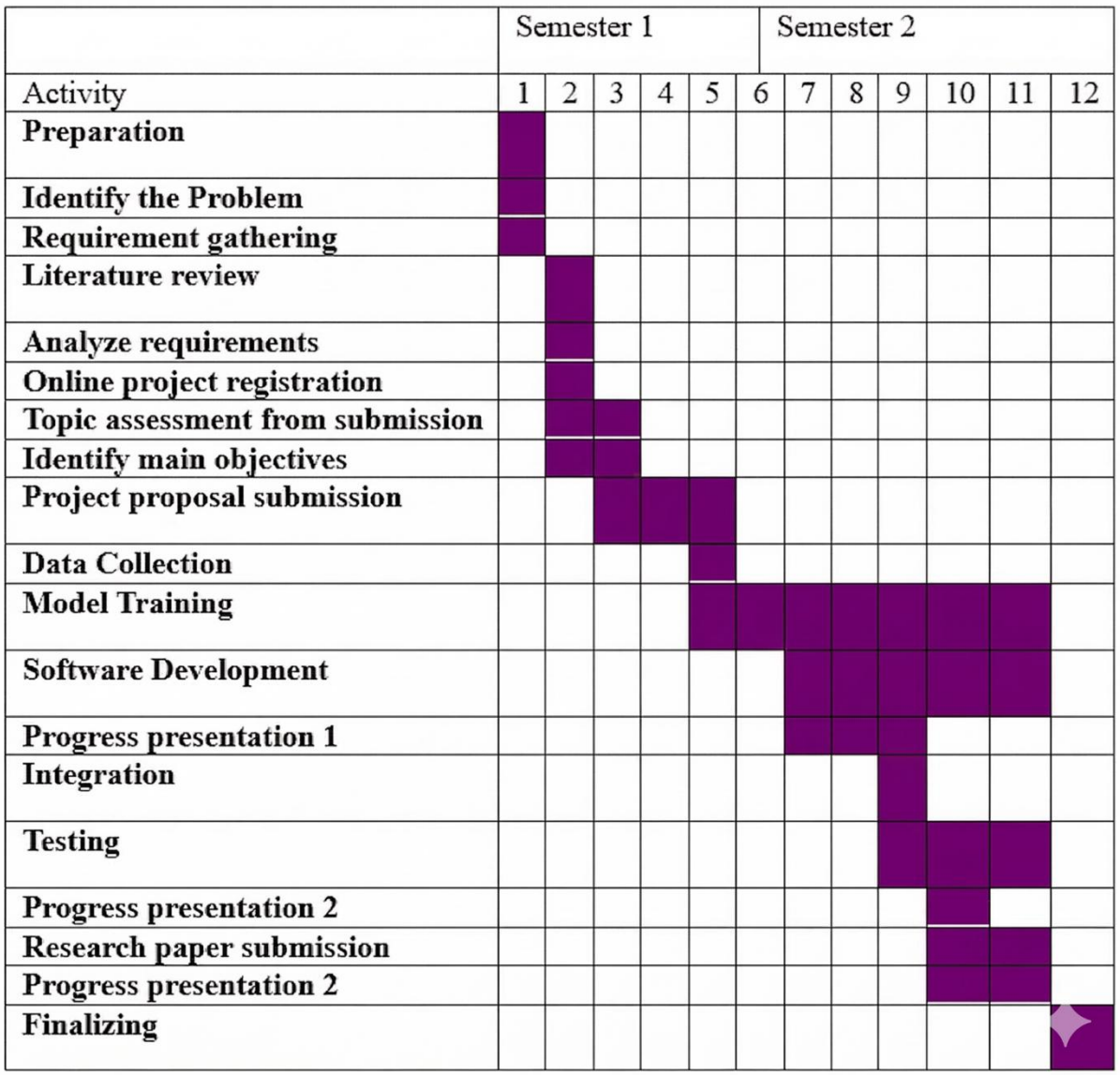

Figure 6 : GANTT CHART

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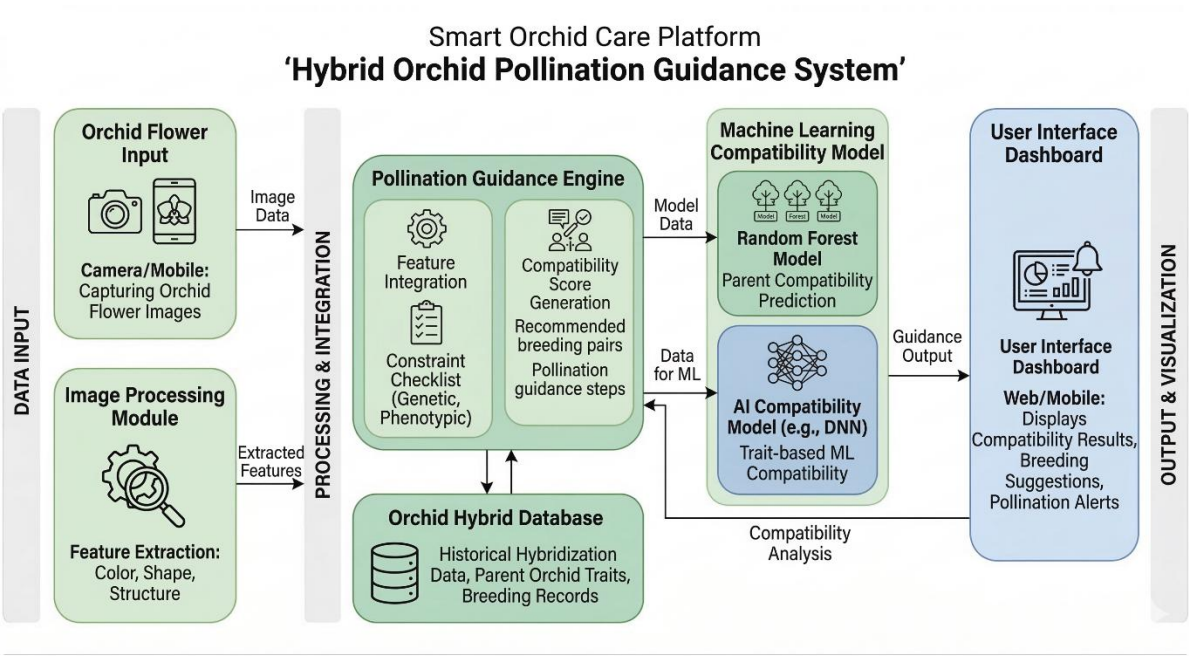
11. APPENDICES

- **Survey instruments**

Appendix A: Orchid Breeder Interview Questions
Topic: Understanding how orchid growers select parent plants for hybrid pollination

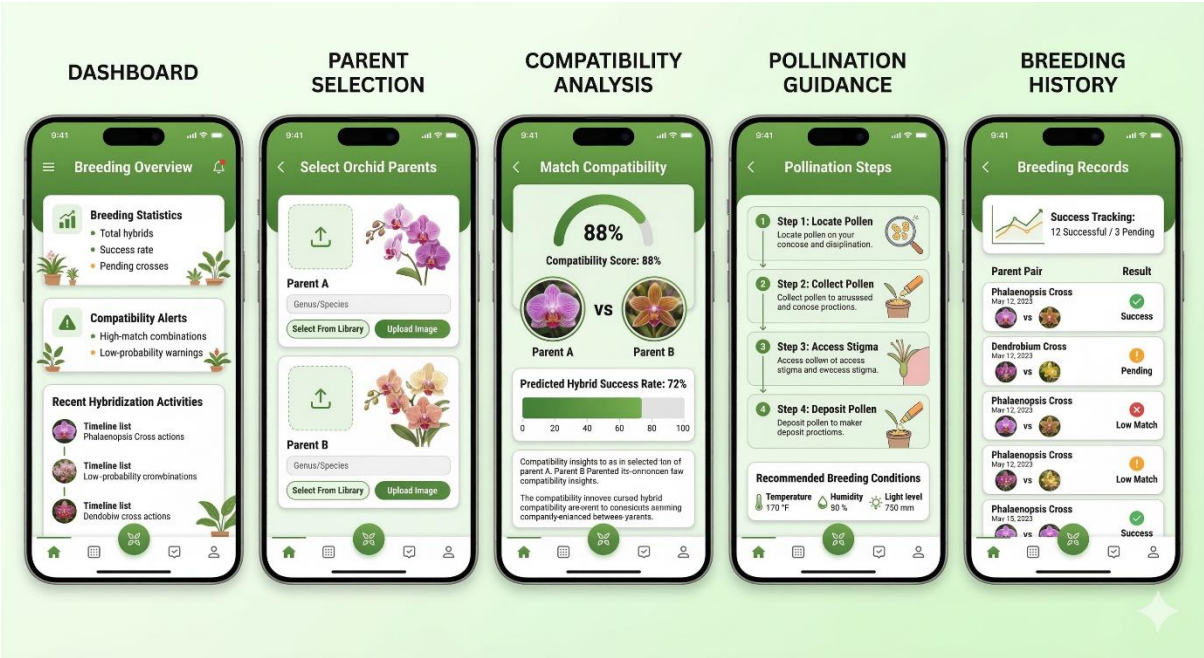
- How do you currently choose orchids for hybrid pollination?
- What characteristics do you consider when selecting parent orchids?
- Do you keep records of previous hybridization results?
- How do you know if two orchids are compatible for breeding?
- Have you experienced unsuccessful pollination attempts?
- What difficulties do you face when predicting hybrid success?
- Would you use a system that analyzes orchid traits and predicts hybrid compatibility?
- How useful would a system that provides pollination guidance and breeding recommendations be for your work?
- What additional features would you expect from a smart orchid breeding support system?

- **Detailed diagrams**



Appendix 1: Detailed System Architecture

- **Preliminary UI mockups**



Appendix 2 : Preliminary UI mockups

- **Risk analysis**

Risk	**Description**	**Mitigation**
Limited dataset	Lack of sufficient orchid images may affect model accuracy	Collect additional images from orchid farms and public datasets
Image quality issues	Poor lighting or blurred images may reduce detection accuracy	Use controlled camera placement and preprocessing techniques
Model prediction errors	ML predictions may not always be fully accurate	Continuous model training and validation
Hardware reliability	Camera or sensor failures may interrupt data collection	Use reliable hardware and backup monitoring methods

Appendix 3 : Risk analysis

- **Ethical documents**

  The proposed research does not involve sensitive human data. However, ethical considerations will still be followed during data collection and system development.

  • Orchid images will be collected only with permission from farm owners.
  • The collected datasets will be used solely for research and system development purposes.
  • No personal or private data from users will be stored or shared.
  • Data security practices will be applied when storing image datasets and prediction results.